

Coexistence and Growth of Prismatic and Basal Dislocation Loops in Irradiated Zirconium

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Abstract

Irradiation growth in hcp zirconium is governed by the partitioning of mobile vacancies and interstitials among crystallographically distinct sinks. Transmission-electron-microscopy studies show abundant prismatic $\langle a \rangle$ loops, predominantly interstitial at lower temperatures with a temperature-dependent fraction of prismatic vacancy $\langle a \rangle$ loops at higher temperatures, together with basal or near-basal vacancy $\langle c \rangle$ -component loops that appear after an incubation dose and drive breakaway growth. This article consolidates the experimental picture, the growth conditions, the symmetry-breaking mechanisms, and a matching reduced cluster-dynamics model into a single consistent account of simultaneous loop growth. We show that Woo-type diffusional-anisotropy-difference (DAD) bias assigns different capture efficiencies to prismatic and basal orientations and thereby opens a finite, non-empty window of the vacancy-to-interstitial mobile-flux ratio over which prismatic interstitial $a_i L$ and basal vacancy $c_v L$ loops grow simultaneously. DAD alone, however, cannot support simultaneous local growth of interstitial and vacancy loops of the same prismatic orientation when their capture coefficients are identical; observed $a_i L$ – $a_v L$ coexistence therefore requires additional physics—production bias from cascade clustering, one-dimensional migration of SIA clusters, variant-resolved prismatic sinks, spatially heterogeneous flux fields, or distinct nucleation barriers. These ingredients act with a clear division of labor: DAD sets the width of the coexistence window, the dislocation and production biases fix where the steady flux ratio sits within it, and fast-interstitial-driven breaking of temporal and spatial symmetry turns the scalar flux ratio into a space-time field that samples both sides of each threshold, removing the same-orientation degeneracy. We realize this account in a compact cluster-dynamics model that treats $a_i L$, $a_v L$, and $c_v L$ as separate evolving populations—with the rare $c_i L$ retained for completeness—whose bilinear net-flux growth laws take their capture efficiencies directly from the Woo DAD coefficients, so that the DAD physics enters as a re-parameterization rather than a structural change and the model reproduces the observed incubation-then-coexistence transient.

1 Introduction: literature context and experimental observations

The irradiation growth of alpha-zirconium and zirconium alloys is a classic manifestation of anisotropic defect accumulation in an hcp lattice. Experiments and reviews consistently associate the dimensional response with the evolution of prismatic $\langle a \rangle$ loops and basal or near-basal $\langle c \rangle$ -component vacancy loops [1–6]. The common loop taxonomy is

$$a_i L : \text{prismatic interstitial } \langle a \rangle \text{ loop}, \quad (1)$$

$$a_v L : \text{prismatic vacancy } \langle a \rangle \text{ loop}, \quad (2)$$

$$c_i L : \text{basal or near-basal interstitial } \langle c \rangle\text{-component loop}, \quad (3)$$

$$c_v L : \text{basal or near-basal vacancy } \langle c \rangle\text{-component loop}. \quad (4)$$

The $\langle a \rangle$ loops have Burgers vector of type

$$\mathbf{b}_a = \frac{1}{3}\langle 11\bar{2}0 \rangle, \quad (5)$$

and are associated with prismatic planes. They may be either interstitial or vacancy in character. The high-dose $\langle c \rangle$ -component loops observed in irradiated Zr are generally vacancy-type and are strongly associated with the transition to breakaway growth [3, 7–11]. Interstitial $\langle c \rangle$ loops are not normally a dominant reported population in neutron-irradiated Zr, but they are retained below as a formal population because the growth inequalities are symmetric once capture coefficients are specified.

The experimentally important complication is the coexistence of interstitial and vacancy prismatic $\langle a \rangle$ loops. The reported vacancy fraction is strongly temperature- and history-dependent. At lower temperatures, especially below roughly 573 K or 300 °C, the $\langle a \rangle$ population is generally dominated by interstitial loops. Near 623 K to 673 K or approximately 350 °C to 400 °C, vacancy $\langle a \rangle$ loops become more significant, with several summaries reporting comparable interstitial and vacancy fractions or a vacancy-rich $\langle a \rangle$ population at the upper end of this range [1–6, 10, 12]. Total prismatic $\langle a \rangle$ loop densities are commonly of order 10^{22} m^{-3} with characteristic diameters of order 5 nm to 20 nm, although individual values depend on temperature, alloy chemistry, dose, dose rate, and imaging method [1, 2, 5, 11–14]. The c_vL population typically appears after an incubation dose and is lower in density but larger in size; for example, $\langle c \rangle$ -loop densities are often reported around 10^{20} – 10^{21} m^{-3} at high dose, with sizes ranging from tens to more than 100 nm depending on temperature and alloy [5, 8, 11, 13].

Table 1 summarizes the qualitative experimental map used in this article. The numerical entries should be read as order-of-magnitude ranges rather than universal constants.

Table 1: Compact map of loop observations in irradiated Zr and Zr alloys. Values combine trends reported in TEM studies and reviews; precise numbers are alloy-, dose-, and imaging-condition dependent.

Temperature regime	Dose regime	Main observations	Representative references
$T \lesssim 573 \text{ K}$	low to several dpa	$\langle a \rangle$ loops form early; mostly interstitial; total $\langle a \rangle$ density commonly $\sim 10^{22} \text{ m}^{-3}$; diameters $\sim 5 \text{ nm}$ to 20 nm .	Jostsons et al. [1], Northwood et al. [2], Holt [3], Adamson et al. [5]
$T \approx 623 \text{ K to } 673 \text{ K}$	low to moderate dpa	Both a_iL and a_vL reported; vacancy fraction increases with temperature and prior deformation/history.	Holt [3], Griffiths [4], Onimus et al. [6], Liu et al. [10], Gilbert et al. [12]
High dose, after incubation	several dpa and above	c_vL loops appear and correlate with breakaway growth; density usually lower than total $\langle a \rangle$ density but size larger.	Holt and Gilbert [7], Tournadre et al. [8], Liu et al. [10], Maron et al. [11], Harte et al. [13]

The theoretical problem is clear: in a homogeneous mean-field model, interstitial and vacancy loops of the same orientation experience the same local mobile-defect fields. If they also have the

same capture efficiencies, one should grow while the other shrinks. Coexistence therefore indicates that at least one idealization is broken: homogeneity, identical capture coefficients, identical nucleation barriers, identical prismatic variants, or identical irradiation history.

2 Loop growth conditions, dislocation bias, and DAD

2.1 Flux scales and net loop growth

Define the mobile interstitial and vacancy flux scales

$$\Phi_I = \sum_{n \in I_{\text{mob}}} n D_{I_n} C_{I_n}, \quad \Phi_V = \sum_{m \in V_{\text{mob}}} m D_{V_m} C_{V_m}, \quad (6)$$

which reduce to $\Phi_I = D_i C_i$ and $\Phi_V = D_v C_v$ in a single-interstitial/single-vacancy approximation. The central local scalar is

$$x \equiv \frac{\Phi_V}{\Phi_I}. \quad (7)$$

This is a flux ratio, not only a concentration ratio.

Let $Z_{s,m}$ be the capture efficiency, or bias factor, of sink family s for mobile species m . In a sink-strength formulation,

$$k_{s,m}^2 = Z_{s,m} \rho_s, \quad (8)$$

where ρ_s is the sink line density or equivalent geometrical sink density. The term *dislocation bias* refers to the fact that $Z_{s,i}$ and $Z_{s,v}$ need not be equal, because point defects have different elastic interactions, transport anisotropies, core interactions, and stress-coupled migration pathways near a dislocation sink [15–18]. Stress-induced preferential absorption (SIPA) and elastodiffusion are stress-dependent contributions to this general bias.

For loop family s and sign χ , where $\chi = +1$ for interstitial loops and $\chi = -1$ for vacancy loops, a compact net driving force is

$$\mathcal{F}_s^{(\chi)} = \chi (Z_{s,I} \Phi_I - Z_{s,V} \Phi_V). \quad (9)$$

Thus an interstitial loop grows if $Z_{s,I} \Phi_I > Z_{s,V} \Phi_V$, while a vacancy loop grows if $Z_{s,V} \Phi_V > Z_{s,I} \Phi_I$. Hence

$$s_i L \text{ grows} \iff x < \frac{Z_{s,I}}{Z_{s,V}}, \quad (10)$$

$$s_v L \text{ grows} \iff x > \frac{Z_{s,I}}{Z_{s,V}}. \quad (11)$$

These inequalities apply to a - and c -oriented loops after choosing the corresponding orientation-resolved capture coefficients. They also show why, under identical local fluxes and identical capture coefficients, $s_i L$ and $s_v L$ of the same orientation cannot both grow simultaneously.

2.2 Woo-type DAD coefficients

Woo–Gösele/Woo DAD theory accounts for the fact that the interstitial diffusion tensor in hcp Zr is more anisotropic than the vacancy tensor [17, 19]. Define

$$p_m = \left(\frac{D_c^m}{D_a^m} \right)^{1/6}, \quad (12)$$

where D_a^m and D_c^m are basal and c -axis diffusion coefficients of mobile species m . For basal-enhanced interstitial diffusion, $D_a^I > D_c^I$ and therefore $p_I < 1$. Vacancies are often approximated as less anisotropic, so $p_V \simeq 1$.

For prismatic $\langle a \rangle$ loop sinks and basal $\langle c \rangle$ loop sinks, commonly used Woo-type orientation factors are

$$Z_{a,m}^{\text{DAD}} = Z_m^0 \frac{p_m + p_m^{-2}}{2}, \quad (13)$$

$$Z_{c,m}^{\text{DAD}} = Z_m^0 p_m. \quad (14)$$

These are reduced orientation-dependent capture factors derived from the more general anisotropic-diffusion sink-bias framework and are used, with related notation, in zirconium irradiation-growth models [11, 17, 20–22].

The prismatic and basal flux thresholds are then

$$x_a^* = \frac{Z_{a,I}^{\text{DAD}}}{Z_{a,V}^{\text{DAD}}} = \frac{Z_I^0 p_I + p_I^{-2}}{Z_V^0 p_V + p_V^{-2}}, \quad (15)$$

$$x_c^* = \frac{Z_{c,I}^{\text{DAD}}}{Z_{c,V}^{\text{DAD}}} = \frac{Z_I^0 p_I}{Z_V^0 p_V}. \quad (16)$$

For nearly isotropic vacancies,

$$x_a^* \simeq \frac{Z_I^0 p_I + p_I^{-2}}{Z_V^0 2}, \quad x_c^* \simeq \frac{Z_I^0 p_I}{Z_V^0}. \quad (17)$$

If $p_I < 1$, then

$$\frac{p_I + p_I^{-2}}{2} > p_I, \quad \Rightarrow \quad x_a^* > x_c^*. \quad (18)$$

This ordering is the basic DAD result: prismatic interstitial loops and basal vacancy loops can both grow when

$$\boxed{x_c^* < x < x_a^*}, \quad (19)$$

or explicitly

$$\boxed{\frac{Z_I^0 p_I}{Z_V^0 p_V} < \frac{\Phi_V}{\Phi_I} < \frac{Z_I^0 p_I + p_I^{-2}}{Z_V^0 p_V + p_V^{-2}}}. \quad (20)$$

With $p_V \simeq 1$ this becomes

$$\boxed{\frac{Z_I^0 p_I}{Z_V^0} < \frac{\Phi_V}{\Phi_I} < \frac{Z_I^0 p_I + p_I^{-2}}{2}}. \quad (21)$$

Numerical estimates at 200°C. In the low-temperature irradiation-growth regime relevant to in-reactor Zr components ($T \simeq 200^\circ\text{C}$), representative values consistent with the models cited above [17, 20–22] are a small net dislocation bias

$$Z_I^0 \simeq 1.1\text{--}1.2, \quad Z_V^0 \simeq 1.0, \quad \frac{Z_I^0}{Z_V^0} \simeq 1.05\text{--}1.15, \quad (22)$$

together with a basal-enhanced SIA diffusion anisotropy $D_c^I/D_a^I \simeq 0.5\text{--}0.6$ and near-isotropic vacancy migration, i.e.

$$p_I = \left(\frac{D_c^I}{D_a^I} \right)^{1/6} \simeq 0.89\text{--}0.92, \quad p_V \simeq 1.0, \quad (23)$$

where the anisotropy ratios follow the atomistic and rate-theory parameterizations of Christien and Barbu [20] and Barashev et al. [21]. Note that D_c^I/D_a^I and hence p_I are only weakly temperature dependent, so these estimates are representative across the ~ 100 – 400°C range.

Flux ratio and critical values at 200°C . The flux ratio driving the loop balance is $x \equiv \Phi_V/\Phi_I$, the ratio of vacancy to interstitial fluxes arriving at the loops. Because Frenkel pairs are produced in equal numbers but interstitials are more mobile and preferentially absorbed at biased sinks, the effective ratio seen by the loop populations is slightly below unity, typically

$$x = \frac{\Phi_V}{\Phi_I} \simeq 0.95\text{--}1.0 \quad (24)$$

in the point-defect balances used here [17, 20, 22]. Substituting the estimates (22)–(23) into the simplified thresholds (17) (with $p_V \simeq 1$) gives

$$x_c^* \simeq \frac{Z_I^0}{Z_V^0} p_I \simeq (1.05\text{--}1.15)(0.89\text{--}0.92) \simeq 0.95\text{--}1.05, \quad (25)$$

$$x_a^* \simeq \frac{Z_I^0}{Z_V^0} \frac{p_I + p_I^{-2}}{2} \simeq (1.05\text{--}1.15)(1.05\text{--}1.08) \simeq 1.10\text{--}1.24. \quad (26)$$

Taking central values $Z_I^0/Z_V^0 \simeq 1.1$ and $p_I \simeq 0.9$ yields $x_c^* \simeq 0.99$ and $x_a^* \simeq 1.17$, so the coexistence window (21) is narrow, $x_c^* \simeq 1.0 < x < x_a^* \simeq 1.2$. The nominal flux ratio $x \simeq 0.95\text{--}1.0$ therefore sits at or just below the lower edge x_c^* , consistent with the observation that simultaneous $a_i L + c_v L$ growth is only marginally favored and highly sensitive to the exact bias and anisotropy parameters.

It is worth stressing that the net dislocation bias and the diffusion anisotropy push the lower threshold x_c^* in opposite directions. The bias factor $Z_I^0/Z_V^0 > 1$ raises x_c^* towards (or above) unity, whereas $p_I < 1$ lowers it. With the “bias-plus-DAD” central values above these two effects nearly cancel, $x_c^* \simeq (1.1)(0.9) \simeq 0.99$, so the coexistence window (21) barely opens. Both biases are physically present—the elastic dislocation bias cannot simply be switched off—so a resolution that relies on discarding one of them is not admissible. The estimate $x \simeq \Phi_V/\Phi_I \simeq 0.95\text{--}1.0$ above, however, was obtained *assuming equal production of freely migrating vacancies and interstitials*; relaxing that assumption is what opens the window.

Under cascade damage the freely migrating point defects are *not* produced in equal numbers: a larger fraction of the interstitials is retained in immobile or one-dimensionally gliding SIA clusters, so the surviving single-defect population reaching the loops is vacancy-rich. Let ϵ_I and ϵ_V denote the fractions of produced interstitials and vacancies that escape the cascade as freely migrating single defects, and define the production-bias factor

$$\eta \equiv \frac{\epsilon_V}{\epsilon_I} \geq 1. \quad (27)$$

Because interstitial clustering is more efficient than vacancy clustering in Zr cascades [17, 21, 23], $\epsilon_I < \epsilon_V$ and hence $\eta > 1$. Since equal-production balances give $\Phi_V/\Phi_I \simeq 1$, the effective flux ratio actually seen by the loops is

$$x = \frac{\Phi_V}{\Phi_I} \simeq \eta x_0, \quad x_0 \simeq 1, \quad (28)$$

where x_0 is the (near-unity) ratio in the absence of production bias. A modest production bias therefore *scales x upward*, and the coexistence condition (21) becomes a condition on η ,

$$\boxed{x_c^* < \eta x_0 < x_a^*} \implies \frac{x_c^*}{x_0} < \eta < \frac{x_a^*}{x_0}. \quad (29)$$

With the central estimates $x_c^* \simeq 0.99$, $x_a^* \simeq 1.17$ and $x_0 \simeq 1$, this requires only

$$\eta \simeq 1.0\text{--}1.2, \quad (30)$$

i.e. a 0–20% excess of freely migrating vacancies over interstitials. Such a production bias is entirely consistent with the free-defect fractions inferred from cascade simulations, so simultaneous $a_iL + c_vL$ growth is recovered with *both* the DAD anisotropy and the dislocation bias retained: it is the production bias η , set by the free-defect fractions, that carries x into the narrow window rather than any single elastic or diffusional bias acting alone.

Crucially, the production bias only *positions* x along the flux-ratio axis; it does not by itself *create* the interval $x_c^* < x < x_a^*$. That interval is opened by the diffusion anisotropy alone. Setting $p_I = p_V = 1$ (no DAD, isotropic capture) in (15)–(16) collapses both thresholds onto the same value,

$$x_c^* = \frac{Z_I^0}{Z_V^0} p_I \xrightarrow{p_I \rightarrow 1} \frac{Z_I^0}{Z_V^0}, \quad x_a^* = \frac{Z_I^0}{Z_V^0} \frac{p_I + p_I^{-2}}{2} \xrightarrow{p_I \rightarrow 1} \frac{Z_I^0}{Z_V^0}, \quad (31)$$

so that $x_a^* = x_c^* = Z_I^0/Z_V^0 \simeq 1.1$ and the window has *zero width*. More generally the window width is set purely by the anisotropy,

$$x_a^* - x_c^* = \frac{Z_I^0}{Z_V^0} \left(\frac{p_I + p_I^{-2}}{2} - p_I \right) = \frac{Z_I^0}{Z_V^0} \frac{1 - p_I^3}{2p_I^2}, \quad (32)$$

which vanishes as $p_I \rightarrow 1$ and is strictly positive only for $p_I < 1$. Hence no production bias can help when DAD is absent: with $x_a^* = x_c^*$ there is no open interval, and multiplying x by any η merely slides the single point ηx_0 from one side of the coincident thresholds to the other, switching between a_iL -only and a_vL/c_vL -only growth without ever satisfying both conditions at once. For example, with $Z_I^0/Z_V^0 \simeq 1.1$ and $p_I = 1$, a production bias $\eta \simeq 1.05$ gives $x \simeq 1.05 < 1.1$, so only a_iL grows, whereas $\eta \simeq 1.15$ gives $x \simeq 1.15 > 1.1$, so only a_vL/c_vL grow; no choice of η yields simultaneous $a_iL + c_vL$ growth. Simultaneous growth therefore requires *both* ingredients: DAD ($p_I < 1$) to open a finite window (32), and a production bias ($\eta > 1$) to place x inside it.

Equation (20) explains simultaneous $a_iL + c_vL$ growth. It does not, by itself, explain simultaneous local $a_iL + a_vL$ growth for identical prismatic capture coefficients.

2.3 General split-threshold condition for a_iL , a_vL , c_iL , and c_vL

For completeness, give each population its own effective capture coefficients:

$$\dot{R}_{aI} = M_{aI} (Z_{aI,I} \Phi_I - Z_{aI,V} \Phi_V), \quad (33)$$

$$\dot{R}_{aV} = M_{aV} (Z_{aV,V} \Phi_V - Z_{aV,I} \Phi_I), \quad (34)$$

$$\dot{R}_{cI} = M_{cI} (Z_{cI,I} \Phi_I - Z_{cI,V} \Phi_V), \quad (35)$$

$$\dot{R}_{cV} = M_{cV} (Z_{cV,V} \Phi_V - Z_{cV,I} \Phi_I). \quad (36)$$

Here M_s includes geometrical conversion factors from absorbed defect content to radius change. The growth conditions are

$$a_i L : \quad x < \frac{Z_{aI,I}}{Z_{aI,V}}, \quad (37)$$

$$a_v L : \quad x > \frac{Z_{aV,I}}{Z_{aV,V}}, \quad (38)$$

$$c_i L : \quad x < \frac{Z_{cI,I}}{Z_{cI,V}}, \quad (39)$$

$$c_v L : \quad x > \frac{Z_{cV,I}}{Z_{cV,V}}. \quad (40)$$

Thus simultaneous growth of $a_i L$, $a_v L$, and $c_v L$ at the same point requires

$$\boxed{\max \left(\frac{Z_{aV,I}}{Z_{aV,V}}, \frac{Z_{cV,I}}{Z_{cV,V}} \right) < \frac{\Phi_V}{\Phi_I} < \frac{Z_{aI,I}}{Z_{aI,V}}}. \quad (41)$$

The interval exists only if

$$\frac{Z_{aI,I}}{Z_{aI,V}} > \max \left(\frac{Z_{aV,I}}{Z_{aV,V}}, \frac{Z_{cV,I}}{Z_{cV,V}} \right). \quad (42)$$

This condition states mathematically that interstitial and vacancy prismatic loops must have different capture physics, nucleation pathways, spatial locations, or histories if they grow simultaneously. If a $c_i L$ population is retained, simultaneous growth of all four formal populations additionally requires

$$\max \left(\frac{Z_{aV,I}}{Z_{aV,V}}, \frac{Z_{cV,I}}{Z_{cV,V}} \right) < x < \min \left(\frac{Z_{aI,I}}{Z_{aI,V}}, \frac{Z_{cI,I}}{Z_{cI,V}} \right). \quad (43)$$

For Zr irradiation growth this four-population condition is mostly formal, because $c_i L$ is not generally the important experimentally observed population.

It is useful to read these formal conditions in light of the two mechanisms identified above. The effective coefficients $Z_{s,I}$ and $Z_{s,V}$ inherit their orientation dependence from the DAD factors p_I, p_V of Eqs. (13)–(14): it is the anisotropy that makes the per-population thresholds $Z_{s,I}/Z_{s,V}$ differ between prismatic and basal sinks, and hence that gives the windows (41) and (43) a finite width. The flux ratio $x = \Phi_V/\Phi_I$ that must fall inside those windows is, in turn, set by the production bias η through (28). In other words, DAD opens the intervals and production bias positions x within them—exactly the division of labor established in (32) and (29) for the prismatic pair.

This reading also exposes the one case that neither mechanism can cover. Simultaneous growth of $a_i L$ and $a_v L$ on the *same* prismatic orientation requires $Z_{aV,I}/Z_{aV,V} < x < Z_{aI,I}/Z_{aI,V}$; but if the two prismatic populations share identical capture coefficients ($Z_{aI,\bullet} = Z_{aV,\bullet}$), these bounds coincide and the interval closes, just as $x_a^* = x_c^*$ when $p_I \rightarrow 1$ in (32). Since DAD assigns a single prismatic factor to *both* interstitial and vacancy $\langle a \rangle$ loops, and production bias only shifts the common x , neither the anisotropy nor η can separate same-orientation prismatic loops. The nonzero interval in (41) for the $a_i L/a_v L$ pair is therefore admissible only if the two populations genuinely differ—in capture physics, nucleation pathway, spatial location, or irradiation history—which is the additional ingredient anticipated by the mechanisms discussed in Section 3.

3 Proposed Mechanism of Simultaneous Loop Growth

3.1 Summary of the settled picture

The preceding analysis fixes three ingredients and their roles. First, the diffusional-anisotropy difference (DAD) between the interstitial and vacancy transport tensors, encoded in the factors $p_I < 1$ and $p_V \simeq 1$ through Eqs. (13)–(14), is what *opens* a finite coexistence window: the prismatic and basal thresholds x_c^* and x_a^* separate only because capture is orientation dependent, and the window width (32) vanishes as $p_I \rightarrow 1$. Second, the elastic dislocation bias $Z_I^0/Z_V^0 > 1$ is always present and merely shifts both thresholds together; it cannot open the window on its own and, if anything, narrows it by pushing x_c^* towards unity. Third, the production bias $\eta = \epsilon_V/\epsilon_I$ of Eqs. (27)–(28), set by the unequal free-defect fractions surviving the cascade, *positions* the flux ratio x within the window.

Taken together these establish co-growth of prismatic interstitial and basal vacancy loops ($a_i L + c_v L$). What they cannot explain is the simultaneous growth of interstitial and vacancy loops of the *same* prismatic orientation ($a_i L + a_v L$): with a single prismatic DAD factor and a single, spatially and temporally uniform x , the two prismatic thresholds coincide and no value of x satisfies both growth conditions at once. A uniform, steady x is therefore too restrictive. The resolution proposed here is that this last assumption fails.

3.2 Symmetry breaking of the mobile-flux ratio

The growth conditions of the split-threshold analysis above were written for a single scalar $x = \Phi_V/\Phi_I$. We now promote x to a field over space and time,

$$x(\mathbf{r}, t) = \frac{\Phi_V(\mathbf{r}, t)}{\Phi_I(\mathbf{r}, t)}, \quad (44)$$

governed by the local, instantaneous fluxes of freely migrating vacancies and interstitials. The central proposal of this paper is that the near-degeneracy of the prismatic thresholds is lifted not by adding capture physics but by *breaking the symmetry* between the two mobile fluxes: whenever $x(\mathbf{r}, t)$ departs from a single value and sweeps across the interval bounded by x_c^* and x_a^* , different loop populations meet their growth conditions at different places or different times. Because interstitials migrate far faster than vacancies ($D_a^I \gg D^V$), the interstitial flux responds first and most sharply to any transient or gradient, so it is naturally the interstitial channel that drives the symmetry breaking. Two limiting realizations follow.

3.3 Temporal symmetry breaking: the nucleation transient

Consider first breaking the symmetry in time. During the initial nucleation phase—which in Zr can extend over of order 1 dpa—the sink structure is sparse and the fast interstitials reach quasi-stationarity long before the slow vacancies. The freely migrating population is therefore strongly interstitial-rich, so that

$$x(t) \ll x_c^* \quad (t \lesssim t_{\text{inc}}), \quad (45)$$

where t_{inc} is the incubation time. In this window every interstitial loop satisfies its growth condition $x < Z_{sI,I}/Z_{sI,V}$ while every vacancy loop is starved: interstitial $\langle a \rangle$ loops grow rapidly, whereas vacancy loops can only nucleate and remain sub-critical or very small. As the microstructure coarsens and the vacancy supersaturation builds up, the fast/slow imbalance relaxes and $x(t)$ rises into the quasi-steady range

$$x(t) \longrightarrow x_{\text{ss}} \in (x_c^*, x_a^*) \quad (t \gtrsim t_{\text{inc}}), \quad (46)$$

at which point the steady-state DAD-plus-production-bias arguments of the previous section apply and vacancy loops also grow. The interstitial loops established during the transient do not disappear; they coexist in the same volume with the vacancy loops that grow later. Simultaneous $a_i L + a_v L$ populations thus arise as a *memory* of the nucleation transient rather than from a single steady flux ratio, and the observed interstitial-loop dominance at low dose followed by the appearance of vacancy loops at higher dose is a direct signature of $x(t)$ crossing the window from below.

3.4 Spatial symmetry breaking: gradient-induced heterogeneity

The same fast interstitial diffusion breaks the symmetry in space. Real specimens contain grain boundaries, free surfaces, precipitates, and dislocation networks that act as defect sinks; near them the mobile-defect concentrations are non-uniform, and because $D_a^I \gg D^V$ the interstitial concentration field is far smoother and reaches much further than the vacancy field. The resulting fluxes therefore have different spatial profiles, and $x(\mathbf{r})$ becomes a genuine function of position. Writing the local balance in terms of the free-defect fractions and the local diffusion lengths $L_m = \sqrt{D_m/k_m^2}$, the flux ratio in the neighborhood of a sink scales as

$$x(\mathbf{r}) \simeq \eta \frac{g_V(\mathbf{r})}{g_I(\mathbf{r})}, \quad (47)$$

where $g_m(\mathbf{r})$ are the geometric flux-weighting functions set by the diffusion lengths. Since $L_I \gg L_V$, interstitials are drained preferentially into the far field while vacancies pile up locally, so $x(\mathbf{r})$ is depressed far from sinks and enhanced close to them. A single specimen can then contain regions where $x(\mathbf{r}) < x_c^*$, in which interstitial loops grow, adjacent to regions where $x(\mathbf{r}) > x_a^*$, in which vacancy loops grow. The gradient in $x(\mathbf{r})$ produced by the interstitial–vacancy mobility asymmetry thus supports spatially heterogeneous, simultaneous growth of both prismatic loop signs without requiring any difference in their intrinsic capture coefficients.

3.5 Synthesis

The proposed mechanism therefore has a clear division of labor. DAD ($p_I < 1$) opens a finite window between x_c^* and x_a^* ; the dislocation and production biases set where the steady-state x sits within it; and the fast-interstitial-driven breaking of the temporal and spatial symmetry of the mobile fluxes turns the single scalar x into a field $x(\mathbf{r}, t)$ that samples *both* sides of each threshold. It is this last step that removes the same-orientation degeneracy: $a_i L$ and $a_v L$ need not grow under identical local, instantaneous conditions, because they grow at different times during the transient (45)–(46) and at different places along the gradient (47). Conventional elastic bias and SIPA remain physically present and modulate the thresholds, but they are modifiers; the essential new ingredient is the space-time variation of x . This picture is testable: it predicts a dose-dependent ordering of loop populations (interstitial loops leading, vacancy loops following) and a spatial correlation of loop sign with proximity to sinks, both of which are accessible to dose-resolved and spatially resolved transmission-electron-microscopy studies.

4 Reduced cluster-dynamics model for loop populations

The mechanism of Sec. 3 is realized numerically by the ZRMICRO reduced cluster-dynamics model, whose state, growth laws, and reaction set are summarized here in the notation of Secs. 2–3.

The model tracks a small set of mobile species together with two families of dislocation loops—interstitial (*a*-type) loops and vacancy (*c*-type) loops—each split into a non-aligned and an aligned sub-population. The growth laws are written directly in terms of the flux scales Φ_I , Φ_V and the orientation- and species-resolved capture efficiencies $Z_{s,m}$ introduced in Eqs. (6)–(11), so that the sign of each loop’s growth is governed by the same flux ratio $x = \Phi_V/\Phi_I$ that controls the split-threshold conditions.

4.1 State variables

Rather than carrying an explicit radius, the model evolves, for each loop family s , a loop *number* concentration C_{sL} and the concentration of point defects *stored* in that family, C_{sL}^c ; the mean radius is then a derived quantity. The full state vector is

$$\mathbf{Q} = \left[\underbrace{C_v, C_i, C_{2i}, C_{3i}}_{\text{mobile}}, \underbrace{C_{iL}, C_{aiL}, C_{vL}, C_{avL}}_{\text{loop numbers}}, \underbrace{C_{iL}^c, C_{aiL}^c, C_{vL}^c, C_{avL}^c}_{\text{loop content}}, \rho_N \right]^T, \quad (48)$$

where C_v is the vacancy concentration and C_i, C_{2i}, C_{3i} are the mono-, di- and tri-interstitial concentrations, the four $C_{\bullet L}$ are loop number densities, the four $C_{\bullet L}^c$ are the corresponding stored-defect contents, and ρ_N is an evolving network dislocation density fed by loop–network coalescence. Subscripts iL, aiL denote non-aligned and aligned interstitial (*a*-type) loops and vL, avL the non-aligned and aligned vacancy (*c*-type) loops. The mean loop radius follows from the stored content and number density,

$$R_{sL} = \ell_s \sqrt{\frac{C_{sL}^c}{C_{sL}}}, \quad s \in \{iL, aiL, vL, avL\}, \quad (49)$$

with ℓ_s a geometric length (ℓ_a for *a*-loops, ℓ_c for *c*-loops) fixed by the atomic volume and loop habit. The loop line density used in the sink strengths is $\rho_{sL} = 2\pi R_{sL} C_{sL}$.

4.2 Mobile species and flux scales

The mobile balances take the standard production–recombination–sink form. Writing G_m for the cascade production of species m , the vacancy and interstitial equations are

$$\dot{C}_v = G_v - \alpha_{iv} C_i C_v - \sum_{n \geq 2} \alpha_{nv} C_n C_v - C_v k_v^2 D_v + (\text{loop release}), \quad (50)$$

$$\dot{C}_i = G_i - \alpha_{iv} C_i C_v - (\text{clustering}) - C_i k_i^2 D_i, \quad (51)$$

with analogous equations for C_{2i} and C_{3i} governed by interstitial clustering and thermal emission. The recombination coefficient scales with the sum of mobilities, $\alpha_{iv} \propto (D_i + D_v)$, and the sink terms use the sink strengths $k_m^2 = \sum_s Z_{s,m} \rho_s$ of Eq. (8). Consistent with Eq. (6), the interstitial flux scale sums over all mobile interstitial sizes weighted by content,

$$\Phi_I = D_i C_i + 2D_{2i} C_{2i} + 3D_{3i} C_{3i}, \quad \Phi_V = D_v C_v, \quad (52)$$

so that the driving flux ratio is $x = \Phi_V/\Phi_I$ exactly as in Eq. (7).

4.3 Loop growth laws and the DAD bias

Loop growth is expressed as the rate of change of stored content, driven by the *net* biased flux of the correct sign for each family. For interstitial (*a*-type) loops the content grows when interstitials

are captured in excess of vacancies, and for vacancy (*c*-type) loops when vacancies are captured in excess of interstitials:

$$\dot{C}_{iL}^c|_{\text{growth}} = \frac{\ell_c}{\ell} \sqrt{C_{iL} C_{iL}^c} (Z_I^a \Phi_I - g Z_V^a \Phi_V), \quad (53)$$

$$\dot{C}_{vL}^c|_{\text{growth}} = \frac{\ell_c}{\ell} Q \sqrt{C_{vL} C_{vL}^c} (Z_V^c \Phi_V - Z_I^c \Phi_I), \quad (54)$$

with identical expressions for the aligned families *aiL*, *avL*. Here $\sqrt{C_{sL} C_{sL}^c}$ is the loop-perimeter prefactor implied by Eq. (49), $g \in [0, 1]$ is a shrinkage gate that halts *a*-loop dissolution below a minimum radius, and Q is the absorption efficiency of basal *c*-loops. Equations (53)–(54) are the content form of the net-flux growth laws of Sec. 2: dividing by the perimeter factor and using (49) recovers $\dot{R}_{sL} \propto Z_{s,V} \Phi_V - Z_{s,I} \Phi_I$, so the growth/shrinkage sign is set by the same inequalities (10)–(11).

The capture efficiencies carry the diffusional-anisotropy difference (DAD) explicitly, and we use the Woo formulation of Sec. 2 directly rather than a phenomenological bias. Identifying the interstitial (*a*-type) loops with prismatic $\langle a \rangle$ sinks and the vacancy (*c*-type) loops with basal $\langle c \rangle$ sinks, the loop capture efficiencies in (53)–(54) are exactly the Woo coefficients (13)–(14),

$$Z_m^a = Z_{a,m}^{\text{DAD}} = Z_m^0 \frac{p_m + p_m^{-2}}{2}, \quad Z_m^c = Z_{c,m}^{\text{DAD}} = Z_m^0 p_m, \quad m \in \{I, V\}, \quad (55)$$

with the anisotropy factors $p_m = (D_c^m/D_a^m)^{1/6}$ of Eq. (12) and the net (dislocation-)bias ratio Z_I^0/Z_V^0 . Substituting (55) into (53)–(54) reproduces the split thresholds of Sec. 2 identically,

$$a_i L \text{ grows} \iff x < \frac{Z_I^a}{Z_V^a} = x_a^*, \quad c_v L \text{ grows} \iff x > \frac{Z_I^c}{Z_V^c} = x_c^*, \quad (56)$$

with x_a^* and x_c^* given by (15)–(16). No change to the solver is required: the growth laws already have the bilinear form $Z_{s,I} \Phi_I - Z_{s,V} \Phi_V$, so adopting the DAD physics amounts to a re-parameterization of the four loop capture efficiencies in (55) rather than a structural change.

For completeness, the phenomenological symmetric split $Z_I^a = 1 + \delta_i$, $Z_I^c = 1 - \delta_i$ (and $Z_V^a = 1 - \delta_v$, $Z_V^c = 1 + \delta_v$) used in earlier versions of the code is recovered from (55) by normalizing each species to its mean *a/c* capture, which gives the exact map

$$\delta_m = \frac{Z_{a,m}^{\text{DAD}} - Z_{c,m}^{\text{DAD}}}{Z_{a,m}^{\text{DAD}} + Z_{c,m}^{\text{DAD}}} = \frac{1 - p_m^3}{1 + 3p_m^3}, \quad (57)$$

so $\delta_m > 0$ whenever $p_m < 1$ and $\delta_V = 0$ for isotropic vacancies ($p_V \simeq 1$). For the representative 200°C anisotropies of Eq. (23) ($p_I \simeq 0.89$ – 0.92 , $p_V \simeq 1.0$), Eq. (57) gives $\delta_i \simeq 0.07$ – 0.09 and $\delta_v \simeq 0$, so a phenomenological run of the code with these values reproduces the DAD split used here; the net bias $Z_I^0/Z_V^0 \simeq 1.05$ – 1.15 of Eq. (22) must still be supplied separately. Equation (57) fixes the anisotropy split; the overall bias Z_I^0/Z_V^0 must be carried separately (it does not appear in δ_m), which is why the DAD coefficients (55) are the more faithful input¹. Since $x_c^* < x_a^*$ for any $p_I < 1$, the two windows overlap: a single steady x in (x_c^*, x_a^*) lets both loop families grow. This is precisely the finite coexistence window of Sec. 3; the model removes the same-orientation degeneracy through the species-resolved DAD capture of (55) rather than by any difference in intrinsic capture geometry.

¹In the **ZrMicro** code these four capture efficiencies are the input keys **Z_i_a**, **Z_v_a**, **Z_i_c**, and **Z_v_c** (for Z_I^a , Z_V^a , Z_I^c , Z_V^c , respectively); setting them from Eq. (55) reproduces the DAD physics without any structural change to the solver.

4.4 Number densities, nucleation, and coalescence

The loop number densities evolve by nucleation, coalescence into the network, and thermal annealing:

$$\dot{C}_{iL} = J_{iL} - K_{iL}^{\text{coal}} C_{iL} (\rho_N + \rho_L), \quad (58)$$

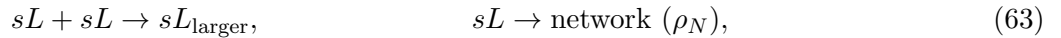
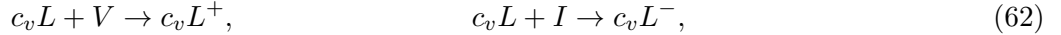
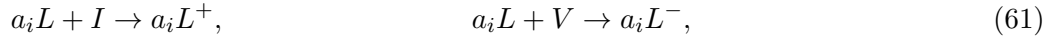
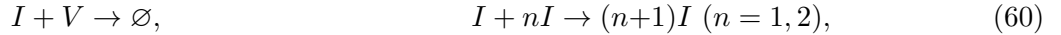
$$\dot{C}_{vL} = J_{vL} - K_{vL}^{\text{coal}} C_{vL} (\rho_N + \rho_L) - \frac{C_{vL}}{\tau_{vL}}, \quad (59)$$

with analogous equations for aiL , avL . The nucleation currents combine a homogeneous clustering source and a cascade source: for interstitial loops $J_{iL} = f_{\text{na}}(R_{i+3i} + R_{2i+2i}) + G_{iL}/n_{iL}^{\text{nuc}}$, while for vacancy loops the cascade term dominates, $J_{vL} = G_{vL}/n_{vL}^{\text{nuc}}$, each nucleus carrying n_{sL}^{nuc} defects that are seeded as content in C_{sL}^c so that nucleation conserves mass. The aligned fraction $f_a = 1 - f_{\text{na}}$ partitions each source between aligned and non-aligned families. Vacancy-loop number density is capped by the annealing term C_{vL}/τ_{vL} , whose released vacancies return to C_v in Eq. (50). Loops absorbed into the network increase ρ_N , which feeds back into the mobile sink strengths.

The transient/steady behavior of Sec. 3 is reproduced by this structure. During the incubation period the fast interstitials give $x(t) \ll x_c^*$, so (53) is strongly positive while (54) is negative and vacancy loops merely nucleate and sit at their content floor; once the vacancy supersaturation builds and $x(t)$ enters (x_c^*, x_a^*) , both (53) and (54) are positive and the two families grow together.

4.5 Reaction set

The reactions implemented in the model are the mobile clustering, recombination, sink-absorption, and loop-exchange channels



where $+/-$ denote content gain/loss, di- and tri-interstitials feed Φ_I through (52), and vacancies and interstitials recombine at rate $\alpha_{iv}C_iC_v$. Opposite-sign loop annihilation $a_iL + c_vL \rightarrow \emptyset$ is available but is used cautiously, since a mean-field term overstates interactions that in reality require proximity, loop mobility, and a favorable elastic geometry.

5 Conclusions and recommendations

This article has assembled the experimental picture, the diffusional-anisotropy-difference (DAD) growth conditions, the symmetry-breaking mechanisms, and a matching reduced cluster-dynamics model into a single consistent account of simultaneous loop growth in irradiated α -zirconium. The main conclusions are as follows.

1. *Experimental setting.* Neutron- and charged-particle data support a temperature- and dose-dependent coexistence of prismatic $\langle a \rangle$ loops (predominantly a_iL at lower temperature, with a growing a_vL fraction at higher temperature) and, after an incubation dose, basal or near-basal vacancy c_vL loops associated with breakaway growth. The total $\langle a \rangle$ population is usually much denser than the high-dose c_vL population, yet the c_vL loops control the breakaway response.

2. *DAD growth window.* Woo-type DAD assigns the prismatic and basal orientations different capture efficiencies, $Z_m^a = Z_m^0(p_m + p_m^{-2})/2$ and $Z_m^c = Z_m^0 p_m$ with $p_m = (D_c^m/D_a^m)^{1/6}$, and thereby opens a finite interval of the mobile-flux ratio $x = \Phi_V/\Phi_I$ over which a_iL and c_vL grow simultaneously,

$$x_c^* = \frac{Z_I^0 p_I}{Z_V^0 p_V} < \frac{\Phi_V}{\Phi_I} < x_a^* = \frac{Z_I^0 p_I + p_I^{-2}}{Z_V^0 p_V + p_V^{-2}}.$$

Because $x_c^* < x_a^*$ for any $p_I < 1$ (with $p_V \simeq 1$), this window is non-empty; for the representative 200°C estimates ($p_I \simeq 0.89\text{--}0.92$, $Z_I^0/Z_V^0 \simeq 1.05\text{--}1.15$) it is comfortably open.

3. *Same-orientation degeneracy.* DAD alone cannot support simultaneous *local, instantaneous* growth of a_iL and a_vL of the same prismatic variant when the two share identical capture coefficients. Observed a_iL – a_vL coexistence therefore signals additional physics: production bias from cascade clustering, one-dimensional migration of SIA clusters, variant-resolved prismatic sinks, spatially heterogeneous flux fields, or distinct nucleation barriers and metastability.
4. *Division of labor among mechanisms.* DAD ($p_I < 1$) sets the width of the coexistence window; the dislocation (net) bias Z_I^0/Z_V^0 and production bias fix where the steady state x sits within it; and the fast-interstitial-driven breaking of temporal symmetry (the nucleation transient) and spatial symmetry (flux gradients) turns the scalar x into a field $x(\mathbf{r}, t)$ that samples both sides of each threshold. This space-time variation, rather than any difference in intrinsic capture geometry, is what removes the same-orientation degeneracy.
5. *Reduced cluster-dynamics model.* The account is realized in a compact cluster-dynamics model that tracks mobile monomers together with di- and tri-interstitials, the four loop number densities $C_{iL}, C_{aiL}, C_{vL}, C_{avL}$ (non-aligned and aligned interstitial a -loops and vacancy c -loops), their stored-defect contents $C_{\bullet L}^c$, and an evolving network density ρ_N . Loop radii are derived from content and number density rather than integrated directly. Growth is written in bilinear net-flux form $Z_{s,I}\Phi_I - Z_{s,V}\Phi_V$, with the capture efficiencies taken *directly* from the Woo DAD coefficients above; substituting them reproduces the split thresholds x_a^*, x_c^* exactly, so the DAD physics enters as a re-parameterization of four capture efficiencies rather than a structural change to the solver. The same coefficients recover the phenomenological split $Z_I^a = 1 + \delta_i$, $Z_I^c = 1 - \delta_i$ (and the vacancy analogues) through the exact map $\delta_m = (1 - p_m^3)/(1 + 3p_m^3)$, giving $\delta_i \simeq 0.07\text{--}0.09$ and $\delta_v \simeq 0$ at 200°C. The model reproduces the incubation-then-coexistence transient: while $x(t) \ll x_c^*$ the interstitial loops grow and the vacancy loops merely nucleate at their content floor, and once $x(t)$ enters (x_c^*, x_a^*) both families grow together.
6. *Recommendations.* For quantitative application the four loop capture efficiencies should be set from the Woo DAD coefficients (with the net bias Z_I^0/Z_V^0 supplied separately, since it does not enter δ_m); temperature-dependent nucleation barriers and vacancy-platelet precursors for c_vL should be retained; and spatially resolved mobile-defect fields should be used where finite boundaries or heterogeneous sinks matter. For as-deformed Zr the deformation-network dislocation density should not be neglected and should be resolved by slip family and, for prismatic systems, by a variant, since the pre-existing dislocation distribution can determine which loop signs and variants grow under irradiation.

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